RAG Application and Project 3

Tuesday, March 17, 2026 6:00 PM

Recap

Built a RAG application

From loading to till generation

1) Indexing

Loaded - Doc loader

Split - Text splitter

{ embed ✓ / stored ✓ } → Chroma.from_documents

2) Retrieved → vectorstore.as_retriever(arg:)

3) Generation → A LCEL chain for the generation ✓

Similarity Search, MMR

Obj only consider 'relevant'

Dis adv. duplicate chunks → remove duplicates

Inc relevance & Inc Diversity ✓

mmr: λ rel − $(1-\lambda)$ diver

$\lambda = 1$ → 100% similarity-search 0%

$\lambda = 0$ → 0% rel 100% Div

λ = b/w (0 1)

Agenda

→ Quickly build another RAG → website data

↳ Creating a chatbot

Creating a chatbot

Project 3: Idea

Youtube → app → Article → web page

Trans → Summarize → Article → webpage

Better prompts

linkedin

Chat with pdf

Created

embed → Sentence Transformer → Docs → emb

Stored emb

Vector db → search for vector

Run Slow

Chromadb

Default chroma create emb using → sentence transformer

× openai

× Gemini emb dis